

SUM-OF-SQUARES AND NON-SUM-OF-SQUARES REGIMES FOR THE TOEPLITZ BÖTTCHER-WENZEL FORM

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ABSTRACT. Let X and Y be real Toeplitz matrices of order N . By the Böttcher–Wenzel inequality, the quartic $2\|X\|^2\|Y\|^2 - 2\langle X, Y \rangle^2 - \|XY - YX\|^2$ (Frobenius norms) in the $4N - 2$ entries of their symbols is nonnegative. For every integer $N \geq 2^{9961475}$ it is not a sum of squares of real quadratic forms, which disproves a conjecture of László (problem MI-15). The negative theorem is formally verified in Lean 4 with Mathlib. The obstruction is an exact negative direction of a limiting tangent kernel, made effective at a finite scale uniformly over all Gram representations. In contrast, the quartic is a sum of squares for every $N \leq 50$.

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1. INTRODUCTION

For a real vector $x = (x_{1-N}, \dots, x_{N-1})$ let $T_N(x) = (x_{i-j})_{0 \leq i, j < N}$ be the real Toeplitz matrix of order N with symbol x . Throughout, norms and inner products of matrices are Frobenius norms and inner products. Put

$$F_N(x, y) = 2\|T_N(x)\|_F^2\|T_N(y)\|_F^2 - 2\langle T_N(x), T_N(y) \rangle_F^2 - \|[T_N(x), T_N(y)]\|_F^2, \quad (1.1)$$

a homogeneous quartic in $4N - 2$ real variables. The Böttcher–Wenzel inequality implies that F_N is nonnegative. László conjectured that this nonnegativity can always be expressed as a sum of squares of real quadratic forms [10, Conjecture 15]; the question is problem MI-15 in the OpenProblemsInNLA catalogue [13]. The following theorem answers it negatively.

Theorem 1.1. *For every integer $N \geq 2^{9961475}$, the polynomial F_N is not a finite sum of squares of real homogeneous quadratic forms. Nevertheless, $F_N(x, y) \geq 0$ for all real symbol vectors x, y and every $N \geq 1$.*

A *sum of squares* (SOS) always means a finite sum of squares of real homogeneous quadratic forms, with arbitrary real coefficients and any number of terms. For a quartic form, requiring the squares to be homogeneous quadratic forms loses nothing: if a form of degree four equals $\sum_j p_j^2$ for real polynomials p_j , then neither the squares of the highest-degree homogeneous parts of the p_j nor those of their lowest-degree parts can cancel, so every p_j is a quadratic form. The nonnegativity assertion is classical (Section 1.3). The threshold is sufficient, not optimized; the first order at which a sum of squares fails is unknown.

1.1. The obstruction in outline. Let $\mathcal{B}_m$ be the restriction of F_{2m} to the m outermost nonzero diagonals on each side, all other symbol variables being zero. After relabelling by depth, the same polynomial is a restriction of F_N for every $N \geq 2m$ (Lemma 3.2). Substitution preserves a sum of squares, so a single obstruction for $\mathcal{B}_m$ transfers to every larger order.

What is tested. By Lemma 3.1, $\mathcal{B}_m$ is a sum of squares exactly when $\mathcal{B}_m = z^T Q z$ for a positive semidefinite Gram matrix Q on the wedge coordinates $z_{ab} = x_a y_b - x_b y_a$; such a Q is determined only up to the Plücker relations among the wedges. The test is a linear functional Φ_m of Q that is nonnegative whenever $Q \succeq 0$. To form it, average Q over two symmetries of the corner, which keeps it positive semidefinite, and split the result into three positive semidefinite blocks. Substituting monomials in two complex variables for the wedge coordinates turns each block into a generating kernel in four complex variables. A fixed combination of these kernels, evaluated at all pairs of 1024 points of $\mathbb{C}^2$, is a positive semidefinite Hermitian 1024×1024 matrix, and Φ_m is its quadratic pairing with a fixed real vector c .

At which points. The points and the coefficients are fixed once and for all:

$$I = \{(r, j, \sigma) : r \in \{1, 2\}, 1 \leq j \leq 256, \sigma \in \{-1, 1\}\}, \quad (1.2)$$

$$a_\ell = r + i\sigma j, \quad b_\ell = r - i\sigma j, \quad c_\ell = \sigma \hat{c}_j, \quad \hat{c}_j = \left\lfloor \frac{2^{43}j}{(128^2 + j^2)^2} \right\rfloor.$$

Thus $|I| = 1024$ and $\|c\|_1 = 859652168 < 2^{30}$. For a dyadic scale $\epsilon = 2^{-k}$ and a corner of depth $m = \epsilon^{-2}$, the point attached to $\ell = (r, j, \sigma)$ is $(ie^{-\epsilon a_\ell}, -ie^{-\epsilon b_\ell})$. It weights a diagonal at depth p by $e^{-\epsilon p r}$ and therefore probes depths of order $1/\epsilon$, deep inside the corner.

Why a negative value contradicts SOS. An exact identity (§2.2) writes the evaluated combination as a part that is the same for every Q minus a single Q -dependent term. After scaling,

the fixed part tends to a limiting tangent kernel whose pairing with c is below -2^{42} by exact rational arithmetic (Lemma 6.1). Positivity of Q forces the Q -dependent term to be small, with constants independent of Q , of its size and of its rank (Sections 4 and 5). Theorem 7.1 states the precise result: for every integer $k \geq 16$ and $m = 2^{2k}$, if $\mathcal{B}_m$ is a sum of squares, then

$$0 \leq \Phi_m < -2^{42} + 2^{90}m^{-1/2} + 2^{80}m^{-1/262144}.$$

The right side is negative once m is large enough. Then no positive semidefinite Gram matrix, and hence no sum of squares, represents $\mathcal{B}_m$.

Corollary 1.2 (Explicit threshold). *The corner $\mathcal{B}_{m_0}$ is not a sum of squares for $m_0 = 2^{9961474}$. Consequently F_N is not a sum of squares for every $N \geq 2m_0 = 2^{9961475}$.*

Proof. Set $k = 38 \cdot 2^{17} + 1 = 4980737$ in the bound (7.2) of Theorem 7.1. Convexity gives $2^{-x} \leq 1 - x/2$ for $0 \leq x \leq 1$, so the right side is

$$-2^{42} + 2^{90-k} + 2^{42}2^{-2^{-17}} \leq -2^{24} + 2^{90-k} < 0.$$

This contradicts $\Phi_{m_0} \geq 0$. Substitution preserves a sum of squares, and the stabilized corner is a restriction of every F_N with $N \geq 2m_0$. $\square$

1.2. Small orders and the open range. For every $1 \leq N \leq 50$, F_N is a sum of squares (Proposition 8.1). Here $F_1 = 0$; for $2 \leq N \leq 50$ the statement follows from exact rational Gram certificates, and for $2 \leq N \leq 20$ it is also formally proved in Lean 4 with Mathlib. The remaining orders are open:

$$51 \leq N < 2^{9961475}.$$

The smallest non-SOS order therefore lies between 51 and $2^{9961475}$, inclusive; monotonicity of the answer within this interval is not known. Section 11 discusses an expectation for this range and the evidence for it.

1.3. Nonnegativity and sums of squares. The Böttcher–Wenzel inequality

$$\|[X, Y]\|_F^2 \leq 2\|X\|_F^2\|Y\|_F^2$$

holds for arbitrary real square matrices [1, 8, 14, 2]. If $X \neq 0$, replacing Y by $Y_\perp = Y - \langle X, Y \rangle_F X / \|X\|_F^2$ does not change the commutator. Expanding $\|Y_\perp\|_F^2$ in the resulting inequality gives exactly $F_N \geq 0$; the case $X = 0$ is immediate. Thus the nonnegativity assertion of Theorem 1.1 holds without the Toeplitz hypothesis.

A nonnegative quartic need not be a sum of squares of quadratic forms. This happens already for biquadratic forms, which have degree two in each of two groups of variables, as F_N has in x and in y : Choi gave a nonnegative biquadratic form in $3 + 3$ variables that is not a sum of squares [3]. A form is a sum of squares exactly when it admits a positive semidefinite Gram matrix, and its Gram matrices form an affine family [4]; this Gram-matrix method underlies both the certificates and the obstruction in this paper. László formulated the Toeplitz question after studying the sum-of-squares problem for the Böttcher–Wenzel form of general matrices [9, 10]; Lu and Wenzel discuss it together with the broader commutator inequalities [12]. The related DDVV inequalities involve sums of commutator norms for families of symmetric matrices [5, 7, 11]. Their sum-of-squares questions are likewise distinct from the inequalities themselves; a recent survey places both in this setting [6].

Theorem 1.1 shows that the Toeplitz structure does not close the gap between the two properties. A sum of squares requires a positive semidefinite Gram matrix on the whole

space of wedge coordinates, whereas nonnegativity only tests decomposable wedges $x \wedge y$. The proof finds a negative direction for the stronger requirement in a limiting kernel and then controls every error at a finite order.

1.4. Verification. The negative conclusion of Theorem 1.1 is formally proved in Lean 4 with Mathlib, for the literal polynomial F_N and arbitrary finite sums of real homogeneous quadratic squares, with no remaining analytic hypothesis. The proof includes the complete corner extraction, the analytic estimates, and a kernel-checked exact witness (Lemma 6.1). Its expanded statement is `ToeplitzSOS.Negative.sharp_negative_explicit`; the negation of the catalogue conjecture is `ToeplitzSOS.Negative.not_MI15`. The positive range is formally proved through order 20. Appendix E gives the declarations, the scope of each formal component, and the relation to the written proof below.

2. THE MECHANISM

The proof follows one line. A sum of squares is a positive semidefinite Gram matrix on wedge coordinates (§2.1). An exact identity removes the freedom of that matrix from one fixed combination of its blocks (§2.2). In a scaling limit, the fixed kernel that remains has a negative direction (§2.3). Positivity makes the only remaining Gram-dependent term small (§2.4), and every error is explicit at a finite scale (§2.5). This section explains each step and why it works; Sections 3–7 give the proofs.

2.1. Wedges and Plücker freedom. For each symbol label λ write $w_\lambda = (x_\lambda, y_\lambda)$ and

$$z_{\lambda\lambda'} = w_\lambda \wedge w_{\lambda'} = x_\lambda y_{\lambda'} - x_{\lambda'} y_\lambda.$$

These wedges vanish simultaneously when the symbol vectors are proportional. Because F_N vanishes when $x = 0$, when $y = 0$, and when $x = y$, every square in a sum of squares vanishes there too, and is therefore a linear combination of the wedges (Lemma 3.1). Hence F_N is SOS exactly when $F_N = z^T Q z$ for a positive semidefinite Gram matrix Q on the wedge space.

Such a Q is far from unique. Wedges of actual symbol vectors obey the Plücker identities

$$z_{ab}z_{cd} - z_{ac}z_{bd} + z_{ad}z_{bc} = 0,$$

and adding a multiple of one of them changes Q without changing the polynomial. A negative eigenvalue of one particular Gram matrix therefore proves nothing: a proof of non-SOS must exclude every Plücker modification at once.

This is also where nonnegativity and sums of squares part ways. Nonnegativity tests $z^T Q z$ only on decomposable vectors $z = x \wedge y$, which satisfy the Plücker identities; a sum of squares requires some Q in the Plücker coset to be positive on all wedge vectors. For large N we construct a linear functional Φ on Gram matrices with two properties: $\Phi(Q) \geq 0$ whenever $Q \succeq 0$, and $\Phi(Q) < 0$ whenever $Q \succeq 0$ represents the corner of F_N . No representation can have both.

2.2. The corner and the cancellation identity. The obstruction lives near the two ends of the Toeplitz symbol. The diagonal at depth p from either end has only $p + 1$ nonzero entries, whatever the ambient order. Diagonals from opposite ends interact in two disjoint corners of the matrix, and the overlap of two such products is the minimum of four depths. Retaining the outer m diagonals on each side therefore gives one polynomial $\mathcal{B}_m$, the same for every $N \geq 2m$, with product weights $(p + 1)(q + 1)$ and a minimum kernel (Lemma 3.2).

The wedges of the corner are *pure* (both diagonals from the same end) or *mixed* (one from each end). Averaging a Gram matrix over a sign change of one side and over the interchange of the two sides loses no feasible matrix. The averaged matrix has no pure–mixed entries, and it splits into two pure blocks $P_{\pm}$, on the sums and differences of the pure wedges of the two sides, and one mixed block M . Each block is positive semidefinite when the Gram matrix is.

Each block is a fixed part plus a correction. The fixed parts come from one explicit Gram matrix of $\mathcal{B}_m$ (Lemma 3.2). Its two pure diagonal blocks both equal D , a diagonal matrix with the product weight $2(p+1)(q+1)$ on the pure wedge of depths $p < q$. Its block coupling the pure wedges of the two sides is K , which pairs wedges with the same gap between their two depths. Its mixed block B consists of the product weights on the diagonal minus a minimum kernel on mixed wedges with equal depth totals.

The corrections carry the entire Plücker freedom. Plücker relations among four diagonals from the same end change the pure diagonal blocks by a *pure four-form* E , a matrix whose entries are totally antisymmetric in the four depths involved. Relations among two diagonals from each end change the block that couples the pure wedges of the two sides by a matrix Θ , symmetric after averaging, and simultaneously change the mixed block by $-\mathcal{R}\Theta$, where $\mathcal{R}$ is the realignment described next. The remaining relations couple pure and mixed wedges, and averaging removes them. Thus

$$P_+ = D + E + K + \Theta, \quad P_- = D + E - K - \Theta, \quad M = B - \mathcal{R}\Theta.$$

To compare pure and mixed blocks, identify each matrix G with its generating kernel $G(\xi, \eta, \zeta, \omega) = f(\xi, \eta)^T G f(\zeta, \omega)$ in four complex variables, where the feature f evaluates a pure wedge of depths $p < q$ as $\xi^p \eta^q - \xi^q \eta^p$ and a mixed wedge as $\xi^p \eta^q$. The *realignment* $\mathcal{R}$ turns a pure kernel P into a mixed kernel by exchanging the two middle variables, $(\mathcal{R}P)(\xi, \eta, \zeta, \omega) = P(\xi, \zeta, \eta, \omega)$. The kernel of a four-form is alternating in all four variables, so $\mathcal{R}E = -E$. Hence both corrections cancel in

$$M + \frac{P_+ + P_-}{2} + \mathcal{R}P_+ = L_m, \tag{2.1}$$

whose right side $L_m = B + D + \mathcal{R}(D + K)$ is fixed. The identity is exact and uses no positivity. If the Gram matrix is positive semidefinite, the first three terms are positive kernels. Realignment does not preserve positive semidefiniteness, so (2.1) does not force L_m to be positive. It reduces the problem to two tasks: find test points on which L_m is negative, and show that $\mathcal{R}P_+$ is small there, uniformly over every allowed Gram matrix.

2.3. The tangent kernel. The test evaluates kernels at nodes $\xi = ie^{-\epsilon a}$, where the speed a has positive real part. Evaluation at ξ weights depth p by $|\xi|^p = e^{-\epsilon p \Re a}$, so these nodes see depths of order $1/\epsilon$, deep inside a corner of depth $m = \epsilon^{-2}$. At this scale the fixed kernel has a limit. After multiplication by ϵ^4 , L_m evaluated at the nodes of the speed pairs (a, b) and (a', b') tends to

$$S_F((a, b), (a', b')) = 2 \left(\frac{2}{A_0^2} + \frac{1}{C_0^2} - \frac{1}{A_0 C_0} \right), \tag{2.2}$$

where $A_0 = (a + \bar{a}')(b + \bar{b}')$ and $C_0 = (a + b)(\bar{a}' + \bar{b}')$.

This tangent kernel has a transparent meaning. For a finite exponential sum $\phi(s, t) = \sum_{\ell} c_{\ell} e^{-a_{\ell}s - b_{\ell}t}$ with real coefficients c_{ℓ} , the pairing $c^* S_{FC}$ equals

$$4 \int_{\mathbb{R}_+^2} st |\phi(s, t)|^2 ds dt + 2 \left| \sum_{\ell} \frac{c_{\ell}}{(a_{\ell} + b_{\ell})^2} \right|^2 - 2 \int_{\mathbb{R}_+^2} |\mathcal{H}\phi(s, t)|^2 ds dt, \quad (2.3)$$

where $\mathcal{H}\phi(s, t) = \int_0^{\infty} \phi(s + u, t + u) du$ integrates ϕ along the diagonal ray starting at (s, t) . The first two terms are positive. The third, a diagonal-tail energy, enters with a negative sign. For the sum given by (1.2), the pairs $\sigma = \pm 1$ combine into multiples of $e^{-r(s+t)} \sin(j(s-t))$, so ϕ is antisymmetric, the middle term vanishes, and the last term exceeds the first. This is the negative direction; Lemma 6.1 certifies it by exact rational arithmetic.

Each exponential alone gives a positive value; only the coherent signed sum is negative. The test therefore acts on Gram matrices, not on values of F_N , which are nonnegative: it contradicts the positive semidefiniteness required of every candidate representation.

2.4. Controlling the realigned term. At the test nodes every entry of $\mathcal{R}P_+$ is a Hermitian diagonal value $P_+[\xi_{\ell}, \bar{\xi}_h]$ of the pure kernel (7.3). For an arbitrary positive semidefinite Gram matrix such values can be of order ϵ^{-4} at nearby complex nodes (Lemma 4.2); after the scaling by ϵ^4 this is the size of the main term. Two facts improve the bound.

First, positivity forces the mixed block to annihilate the complete antidiagonal sums of low total (Lemma 4.1). This follows by evaluating all retained variables on a geometric progression and comparing the first nonzero coefficient of a sum of real squares. Incomplete high antidiagonals are not forced to vanish; their effect is exponentially small at the nodes used here (Lemma 4.3). As a result, on the conjugate set $\eta = \bar{\xi}$ the pure kernel is at most of order ϵ^{-2} , with an explicit finite tail.

Second, a finite positive semidefinite matrix P_+ is the Gram matrix of a holomorphic vector V with $\|V(\xi, \eta)\|^2 = P_+[\xi, \eta]$, and $\log \|V\|^2$ is subharmonic. Two maximum-principle comparisons on rectangles, one in each speed variable, interpolate between the bounds ϵ^{-4} and ϵ^{-2} and give

$$\epsilon^4 \|V\|^2 \leq 2^{20} \epsilon^{2^{-17}}$$

on the region needed by all 1024^2 entries of the test (Lemma 5.1). The exponent is small because each comparison retains only a lower bound for the harmonic weight of one side of a rectangle. The estimate holds for finite matrices of every size and rank.

2.5. The finite estimate and the size of the threshold. With $m = \epsilon^{-2}$, the finite kernel $\epsilon^4 L_m$ differs from S_F by at most $2^{30} \epsilon$ per entry at the test nodes, and $\|c\|_1^2 < 2^{60}$. Pairing (2.1) with c gives

$$0 \leq c^* G_m c = \epsilon^4 c^* \mathbf{L}_m c - \epsilon^4 c^* \mathbf{R}_m c < -2^{42} + 2^{90} \epsilon + 2^{80} \epsilon^{2^{-17}},$$

where G_m , $\mathbf{L}_m$ and $\mathbf{R}_m$ are the matrices of evaluated entries of $\epsilon^4(M + (P_+ + P_-)/2)$, L_m and $\mathcal{R}P_+$; the first is positive semidefinite. This is Theorem 7.1.

The threshold is large for a quantitative reason. The integer coefficients make the main term look large, $c^* S_{FC} < -2^{42}$, but every entrywise error is multiplied by $\|c\|_1^2$, and relative to $\|c\|_1^2$ the certified negativity is small: $-c^* S_{FC} / \|c\|_1^2$ lies between 2^{-18} and 2^{-16} , on the scale of 10^{-5} . The error $2^{90} \epsilon$ is harmless, but $2^{80} \epsilon^{2^{-17}}$ decays extremely slowly and falls below 2^{42} only when $\epsilon < 2^{-38 \cdot 2^{17}}$. The largest dyadic ϵ with a negative budget is $2^{-4980737}$, which gives

$m = \epsilon^{-2} = 2^{9961474}$ (Appendix C). These constants explain the sufficient threshold; they do not predict the location of the first non-SOS order.

2.6. Why small orders are sums of squares. At a fixed small order the Plücker freedom can make a Gram matrix positive on the complement of its forced kernel. The exact certificates through order 50 do precisely this, with strictly positive rational bounds on the relevant compressed blocks (Section 8). The analytic obstruction requires a deep corner and a geometric decay length $1/\epsilon$ that fits within it, at a scale where the errors above fall below the main term; no estimate here makes it effective at small orders. Positive certificates must nevertheless lose their relative margin as the order grows (Section 9), and the natural induction through positive increments fails at all large orders (Section 10).

3. THE WEDGE FRAME AND THE STABILIZED CORNER

3.1. The complete quadratic frame. For real vectors x, y indexed by an ordered set of labels, write $z_{ab} = x_a y_b - x_b y_a$ for labels $a < b$.

Lemma 3.1 (The complete quadratic frame). *Let $F(x, y)$ be a real homogeneous quartic that vanishes whenever $x = 0$, $y = 0$, or $y = x$. If F is a sum of squares of real homogeneous quadratics, then every square is a linear combination of the z_{ab} . Moreover the kernel of the map $Q \mapsto z^T Q z$ on symmetric matrices is exactly the span of the Plücker matrices*

$$z^T P_{abcd} z = 2(z_{ab} z_{cd} - z_{ac} z_{bd} + z_{ad} z_{bc}), \quad a < b < c < d.$$

In particular, if $F = z^T G z$, then

$$F \text{ is SOS} \iff G + E \succeq 0 \text{ for some } E \in \text{span}\{P_{abcd}\}. \quad (3.1)$$

Proof. Write $F = \sum_j q_j^2$. At each point with $y = 0$, the sum of real squares is zero; hence every $q_j(x, 0)$ is identically zero. The same holds at $x = 0$. A homogeneous quadratic thus contains only terms $x_a y_b$, so $q_j = x^T A_j y$. Now $q_j(x, x) = 0$ for all real x implies $A_j + A_j^T = 0$. Therefore $q_j = \sum_{a < b} (A_j)_{ab} z_{ab}$. The coefficient vectors of these quadratics give a positive semidefinite Gram matrix.

For the relation space, group products of wedges according to their multiset of labels. Polynomial monomials from different multisets cannot cancel. A multiset with a repeated label has at most one nonzero pairing, up to the antisymmetric signs; its wedge product is a nonzero polynomial. On four distinct labels there are precisely the three products $z_{ab} z_{cd}$, $z_{ac} z_{bd}$, $z_{ad} z_{bc}$. Their displayed alternating combination is zero. Any two are linearly independent: for example their coefficients of $x_a x_c y_b y_d$ and $x_a x_b y_c y_d$ already have rank two. Thus their relation space is one-dimensional, generated by the Plücker identity. This proves the complete kernel description.

If an SOS Gram represents the same polynomial as G , their difference belongs to this kernel. Conversely a positive semidefinite representative has a real Gram factorization, whose rows are the required wedge quadratics. This proves (3.1). $\square$

The form F_N and each of its restrictions vanish when $x = 0$, $y = 0$ or $x = y$, so Lemma 3.1 applies to them. Positivity in (3.1) is required on all of wedge space, not only on the decomposable vectors $x \wedge y$; this is the gap described in §2.1.

3.2. Coordinates and conventions. Let $N \geq 2m$. For $0 \leq p < m$ the retained symbol pairs at depth p are

$$w_p^- = w_{\lambda_p^-}, \quad w_p^+ = w_{\lambda_p^+}, \quad \lambda_p^\pm = \pm(N-1-p),$$

where $w_\lambda = (x_\lambda, y_\lambda)$; the diagonal at depth p from either end has $p+1$ entries. Their wedges are the pure wedges $u_{pq} = w_p^- \wedge w_q^-$ and $v_{pq} = w_p^+ \wedge w_q^+$ for $p < q$, and the mixed wedges $z_{pq}^{\text{mix}} = w_p^- \wedge w_q^+$ for all p, q . The notation of the analytic argument is collected here.

Symbol	Meaning
$w_\lambda; w_p^\pm$	symbol pair at label λ ; retained pair at depth p
$u_{pq}, v_{pq}; z_{pq}^{\text{mix}}$	pure wedges of the two sides; mixed wedges
D, K, B	fixed pure diagonal, pure cross and mixed matrices (3.4)
$E, \Theta; \Psi = M - B$	pure four-form, pure cross correction; mixed correction
$P_\pm, M$	averaged pure and mixed blocks (3.6)
$\mathcal{R}; L_m; S_F$	realignment (3.2); fixed corner kernel (3.9); tangent kernel (6.1)
ξ, η, ζ, ω	complex kernel arguments
$f_{pq}(\xi, \eta); \mathbf{e}_{pq}$	pure feature; mixed basis vector
$A, C, \Delta; \Pi$	kernel denominators; product of the four arguments (Section 4)
$k, \epsilon = 2^{-k}, m = \epsilon^{-2}$	dyadic exponent, scale and corner depth
$a_\ell, b_\ell; c_\ell; \hat{c}_j$	witness speeds; witness coefficients; their integer weights (1.2)
$G_m; \Phi_m$	evaluated Gram matrix; its pairing with c (7.1)
$\alpha, \beta; \mathcal{H}$	continuation coordinates (Section 5); diagonal-ray integral (2.3)

Kernel conventions. A Gram matrix G represents the polynomial $q^T G q$, so a symmetric off-diagonal entry G_{ij} contributes $2G_{ij}q_i q_j$; every matrix below is normalized this way. Pure wedges are evaluated by the feature $f_{pq}(\xi, \eta) = \xi^p \eta^q - \xi^q \eta^p$, mixed wedges by $\xi^p \eta^q$. A matrix G is identified with its four-slot generating kernel $G(\xi, \eta, \zeta, \omega) = f(\xi, \eta)^T G f(\zeta, \omega)$, formed with the appropriate features, and $P[\xi, \eta] = P(\xi, \eta, \bar{\xi}, \bar{\eta})$ denotes the Hermitian diagonal of a kernel P . Sums of pure and mixed matrices are sums of their kernels. A subscript m denotes a finite depth- m kernel, and a subscript ∞ its absolutely convergent infinite series.

Realignment. For a pure kernel P , the mixed kernel $\mathcal{R}P$ is

$$(\mathcal{R}P)(\xi, \eta, \zeta, \omega) = P(\xi, \zeta, \eta, \omega). \quad (3.2)$$

In entries, extend a symmetric pure matrix Ξ antisymmetrically in each index pair, $\Xi_{qp,rs} = \Xi_{pq,rs} = -\Xi_{pq,rs}$ and $\Xi_{pp,rs} = 0$. Then $\mathcal{R}\Xi$ is the symmetric mixed matrix with

$$\mathcal{R}(\Xi)_{pr,qs} = \Xi_{pq,rs}, \quad \mathcal{R}(\Xi)_{ps,qr} = -\Xi_{pq,rs} \quad (p < q, r < s), \quad (3.3)$$

and every other entry is a symmetric counterpart of these or zero. Indeed $\Xi(\xi, \zeta, \eta, \omega) = \sum_{p < q, r < s} \Xi_{pq,rs} (\xi^p \zeta^q - \xi^q \zeta^p) (\eta^r \omega^s - \eta^s \omega^r)$, and the mixed monomials $\xi^p \eta^r \zeta^q \omega^s$ and $\xi^p \eta^s \zeta^q \omega^r$ carry the entries (pr, qs) and (ps, qr) .

A pure four-form is a pure matrix $E_{pq,rs} = \Omega_{pqrs}$ ($p < q, r < s$) with Ω totally antisymmetric in its four indices; these are exactly the combinations of the pure Plücker matrices of Lemma 3.1. Its kernel $E(\xi, \eta, \zeta, \omega) = \sum_{p,q,r,s} \Omega_{pqrs} \xi^p \eta^q \zeta^r \omega^s$ is alternating in its four arguments. Hence $\mathcal{R}E = -E$, and E vanishes when two arguments coincide.

3.3. The normal form. By Lemma 3.1, the Gram matrices of the corner form a coset of the Plücker span. Two symmetries of the corner reduce this coset to two corrections.

Lemma 3.2 (Stabilization and the complete normal form). *Let $N \geq 2m$. The restriction of F_N to the retained pairs $w_p^\pm$, $0 \leq p < m$, all other symbol variables being zero, is a polynomial $\mathcal{B}_m$ that does not depend on N . In the ordered wedge coordinates (u, v, z^{mix}) it is represented by the Gram matrix with pure block $\begin{pmatrix} D & K \\ K & D \end{pmatrix}$, mixed block B , and no pure-mixed entries, where*

$$\begin{aligned} D_{pq,rs} &= 2(p+1)(q+1)\delta_{pr}\delta_{qs}, \quad p < q, \quad r < s, \\ K_{pq,rs} &= -2\mathbf{1}_{q-p=s-r} \min(p+1, r+1), \quad p < q, \quad r < s, \\ B_{pq,rs} &= 2(p+1)(q+1)\delta_{pr}\delta_{qs} - 2\mathbf{1}_{p+q=r+s} \min(p+1, q+1, r+1, s+1). \end{aligned} \quad (3.4)$$

Every positive semidefinite Gram matrix representing $\mathcal{B}_m$ becomes, after averaging over the side sign change $w^- \mapsto -w^-$ and the side interchange $w^- \leftrightarrow w^+$,

$$Q_{\text{av}} = \begin{pmatrix} D+E & K+\Theta & 0 \\ K+\Theta & D+E & 0 \\ 0 & 0 & B-\mathcal{R}\Theta \end{pmatrix} \succeq 0, \quad (3.5)$$

where E is a pure four-form and Θ an unrestricted real symmetric matrix on the pure wedge space. The orthogonal change to $(u+v)/\sqrt{2}$, $(u-v)/\sqrt{2}$, z^{mix} gives the positive semidefinite blocks

$$P_+ = D + E + K + \Theta, \quad P_- = D + E - K - \Theta, \quad M = B - \mathcal{R}\Theta. \quad (3.6)$$

In particular $\mathcal{R}E = -E$.

Proof. Baseline. Write $T_N(x) = \sum_\lambda x_\lambda S_\lambda$ with the shift matrices S_λ . Distinct shifts are Frobenius orthogonal, and $\|S_\lambda\|_F^2 = N - |\lambda|$, which is $p+1$ at depth p . The weighted Lagrange identity for the first two terms of (1.1) and bilinearity of the commutator give

$$F_N = \sum_{\lambda < \lambda'} 2(N - |\lambda|)(N - |\lambda'|) z_{\lambda\lambda'}^2 - \left\| \sum_{\lambda < \lambda'} z_{\lambda\lambda'} [S_\lambda, S_{\lambda'}] \right\|_F^2.$$

On the retained pairs, same-side shifts commute, so only mixed wedges enter the commutator term, and direct multiplication gives

$$\langle [S_{\lambda_p^-}, S_{\lambda_q^+}], [S_{\lambda_r^-}, S_{\lambda_s^+}] \rangle_F = 2\mathbf{1}_{p-q=r-s} \min(p+1, q+1, r+1, s+1). \quad (3.7)$$

Each such commutator is supported on two strings of entries near opposite corners of the matrix; for $N \geq 2m$ the strings are disjoint and independent of N . This proves the stabilization. The resulting raw Gram matrix has the diagonal of D and B , and its only off-diagonal entries join mixed wedges $z_{pq}^{\text{mix}}, z_{rs}^{\text{mix}}$ with $p-q=r-s$. For such a pair with $p < r$ and $q < s$, the entry $-2 \min(p+1, q+1)$ contributes $-4 \min(p+1, q+1) z_{pq}^{\text{mix}} z_{rs}^{\text{mix}}$. The Plücker relation

$$u_{pr}v_{qs} - z_{pq}^{\text{mix}} z_{rs}^{\text{mix}} + z_{ps}^{\text{mix}} z_{rq}^{\text{mix}} = 0 \quad (3.8)$$

rewrites this contribution as $-4 \min(p+1, q+1)(u_{pr}v_{qs} + z_{ps}^{\text{mix}} z_{rq}^{\text{mix}})$: a pure pair with equal gaps $r-p=s-q$ and a mixed pair with equal totals $p+s=r+q$. Summing over all raw pairs gives exactly K and B in (3.4).

Freedom. By Lemma 3.1, any other Gram matrix of $\mathcal{B}_m$ differs from this one by a combination of Plücker matrices on four retained pairs. Four pairs from one side give a pure relation in u or in v ; three pairs from one side give a relation coupling pure and mixed wedges; two pairs from each side give (3.8) up to relabelling. Adding 2θ times (3.8) adds θ to the cross entry (u_{pr}, v_{qs}) , and $-\theta$ and θ to the mixed entries $(z_{pq}^{\text{mix}}, z_{rs}^{\text{mix}})$ and $(z_{ps}^{\text{mix}}, z_{rq}^{\text{mix}})$. By (3.3), a cross correction Θ thus comes with the mixed correction $-\mathcal{R}\Theta$.

Averaging. The side sign change preserves $\mathcal{B}_m$, because the baseline has no pure–mixed entries; the side interchange transposes both Toeplitz matrices and preserves (1.1). Conjugating a positive semidefinite representing Gram matrix by either map gives another one. The sign change fixes u, v and negates z^{mix} , so averaging over it removes all pure–mixed couplings. The interchange maps $u_{pq} \mapsto v_{pq}$ and $z_{pq}^{\text{mix}} \mapsto -z_{qp}^{\text{mix}}$, so averaging over it makes the two pure diagonal blocks equal, with a common four-form E , and the cross block symmetric. This gives (3.5); the orthogonal change of coordinates gives (3.6). $\square$

Set $Q_0 = D + K$. An exact identity now removes both arbitrary corrections:

$$M + (P_+ + P_-)/2 + \mathcal{R}P_+ = B + D + \mathcal{R}Q_0 =: L_m. \quad (3.9)$$

Indeed the correction terms are $-\mathcal{R}\Theta + E + \mathcal{R}E + \mathcal{R}\Theta = 0$. This algebra uses no positivity. Positivity is used only for the first three terms, which are kernels of positive semidefinite matrices; the realigned term need not be positive. This is the identity (2.1).

4. FORCED MEANS AND FINITE TAILS

This section supplies the bounds that hold for every positive semidefinite Gram matrix of the corner, whatever its corrections: the forced low means, the generating kernels of the fixed baselines, a global bound for the pure blocks, and explicit finite tails.

Lemma 4.1 (Forced complete means). *Let $s_g = \sum_{p+q=g} \mathbf{e}_{pq}$ in the mixed coordinates. Then $B \succeq 0$, and every PSD Gram (3.6) satisfies*

$$Ms_g = Bs_g = 0 \quad (0 \leq g < m). \quad (4.1)$$

Proof. The matrix B is block diagonal by totals $g = p + q$. On a complete total g , that is, when all $p, q \geq 0$ with $p + q = g$ are retained, put $d = \min(p + 1, g - p + 1)$. Then

$$\sum_{r=0}^g \min(d, r + 1, g - r + 1) = \sum_{h=1}^d (g + 3 - 2h) = d(g + 2 - d) = (p + 1)(g - p + 1),$$

so every row of the block sums to zero. Its off-diagonal entries are nonpositive, so it is a weighted graph Laplacian, positive semidefinite with s_g in its kernel. A block with $g \geq m$ is a principal submatrix of the complete block given by the same formula. Hence $B \succeq 0$, and $Bs_g = 0$ for $g < m$, since these totals are complete.

Now let $Q \succeq 0$ represent $\mathcal{B}_m$, and substitute $w_p^- = (t^p, 0)$, $w_p^+ = (0, t^p)$. The pure wedges vanish and $z_{pq}^{\text{mix}} = t^{p+q}$, so the substituted polynomial is $\|\sum_g t^g Q^{1/2} s_g\|^2$, with s_g placed in the mixed coordinates. Computed from the baseline, it equals $\sum_{g,h} t^{g+h} s_g^T B s_h$, which has no term of degree below $2m$. If $Q^{1/2} s_g \neq 0$ for some $g < m$, the least such g makes the coefficient of t^{2g} equal to $\|Q^{1/2} s_g\|^2 > 0$, a contradiction. Hence $Qs_g = 0$; in particular the mixed block satisfies $Ms_g = 0$. This proves (4.1). $\square$

Only these *complete low means* are forced; the means of the incomplete totals $m \leq g \leq 2m - 2$ need not vanish. Symmetry and (4.1) imply that $(M - B)(\xi, \xi, \zeta, \zeta)$ contains only terms $\xi^g \zeta^h$ with both $g, h \geq m$. They will be bounded as finite tails.

To sum the fixed baselines, put

$$A = (1 - \xi\zeta)(1 - \eta\omega), \quad \Delta = (1 - \xi\omega)(1 - \eta\zeta), \quad C = (1 - \xi\eta)(1 - \zeta\omega), \quad \Pi = \xi\eta\zeta\omega.$$

Absolute geometric series in the open polydisk give

$$\begin{aligned} D_\infty &= 2(A^{-2} - \Delta^{-2}), \\ B_\infty &= 2(A^{-2} - (A\Delta)^{-1}), \\ Q_{0,\infty} &= D_\infty - 2(\Delta - A)/(A\Delta C), \\ L_\infty &= 2\{2/A^2 + 1/C^2 - 1/(AC) - 2/\Delta^2\}. \end{aligned} \tag{4.2}$$

Appendix A derives these sums.

Lemma 4.2 (Global bound and the repeated-node identity). *For every feasible finite averaged positive semidefinite Gram and $|\xi|, |\eta| < 1$,*

$$0 \leq P_\pm[\xi, \eta] \leq \frac{8}{(1 - |\xi|^2)^2(1 - |\eta|^2)^2}. \tag{4.3}$$

Writing $\Psi = M - B$, one also has

$$P_+[\xi, \bar{\xi}] = Q_{0,m}[\xi, \bar{\xi}] + \Psi(\xi, \xi, \bar{\xi}, \bar{\xi}). \tag{4.4}$$

For $\xi = \sqrt{\rho}e^{i\vartheta}$, the infinite baseline on this diagonal is

$$Q_{0,\infty}[\xi, \bar{\xi}] = \frac{8\rho \sin^2 \vartheta}{(1 - \rho)^2((1 - \rho)^2 + 4\rho \sin^2 \vartheta)^2}. \tag{4.5}$$

Proof. Alternation of E , before using positivity, gives

$$P_+[\xi, \eta] + P_-[\xi, \eta] + P_+[\xi, \bar{\eta}] + P_-[\xi, \bar{\eta}] = 2D_m[\xi, \eta] + 2D_m[\xi, \bar{\eta}]. \tag{4.6}$$

Every term on the left is nonnegative. The diagonal D_m is bounded by its positive infinite sum, at most $2/((1 - |\xi|^2)^2(1 - |\eta|^2)^2)$, proving (4.3). On the repeated slots $(\xi, \bar{\xi}, \bar{\xi}, \xi)$, E vanishes and

$$\Psi(\xi, \xi, \bar{\xi}, \bar{\xi}) = -\Theta(\xi, \bar{\xi}, \xi, \bar{\xi}) = \Theta(\xi, \bar{\xi}, \bar{\xi}, \xi),$$

which proves the plus sign in (4.4). Both (4.4) and (4.6) are algebraic identities for every averaged Gram; forced means enter only when bounding the correction in (4.4). Substitution in (4.2), with $A = C = (1 - \rho)^2$ and $\Delta = A + 4\rho \sin^2 \vartheta$, gives (4.5). $\square$

Lemma 4.3 (Uniform finite tails). *For $0 < \epsilon \leq 1$, set*

$$\mathcal{E}(m, \epsilon) = 2^{32}\epsilon^{-8}e^{-\epsilon m/4}. \tag{4.7}$$

If all four slot moduli are at most $e^{-\epsilon/2}$, then $|L_m - L_\infty| \leq \mathcal{E}$. For every averaged positive semidefinite Gram, also $|P_+[\xi, \bar{\xi}] - Q_{0,\infty}[\xi, \bar{\xi}]| \leq \mathcal{E}$ when $|\xi| \leq e^{-\epsilon/2}$.

Proof. Antisymmetry gives $\text{diag}(\mathcal{R}\Theta) = 0$. Thus the PSD matrices M, B have the same diagonal, bounded by $2(p+1)(q+1)$. Cauchy–Schwarz yields, with $W = (p+1)(q+1)(r+1)(s+1)$,

$$|\Psi_{pq,rs}| \leq 4\sqrt{W} \leq 4W. \tag{4.8}$$

Every expanded coefficient of $Q_{0,\infty}$ and L_∞ has absolute value at most $16W$: the diagonal and minimum terms in (3.4) give bounds $2W$ for D , $2W$ for K , and $4W$ for B ; even the triangle bound $10W$ suffices for L_∞ . In a discarded term at least one index is at least m . Split its radial decay in half, choose this index in four ways, and take the product of four copies of $\sum_{j \geq 0} (j+1)e^{-\epsilon j/4} = (1 - e^{-\epsilon/4})^{-2}$. Since $1 - e^{-\epsilon/4} \geq \epsilon/8$, the truncation error is at most

$$64e^{-\epsilon m/4}(1 - e^{-\epsilon/4})^{-8} \leq 2^{30}\epsilon^{-8}e^{-\epsilon m/4}. \tag{4.9}$$

The repeated correction in (4.4) has both total degrees at least m by Lemma 4.1. Using (4.8) in the same sum bounds it by $2^{26}\epsilon^{-8}e^{-\epsilon m/2}$. Combining these bounds proves (4.7), uniformly over every allowed E, Θ . $\square$

For $k \geq 16$, $\epsilon = 2^{-k}$, $m = \epsilon^{-2}$, the elementary inequality $2^{k-2} \geq 8k + 40$ follows by checking $k = 16$ and doubling. Using $e > 2$ gives

$$\mathcal{E} \leq 2^{32+8k-2^{k-2}}, \quad \mathcal{E} \leq \epsilon^{-2}, \quad \epsilon^4 \mathcal{E} \leq \epsilon. \quad (4.10)$$

The last two claims require respectively $2^{k-2} \geq 6k + 32$ and $2^{k-2} \geq 5k + 32$, both consequences of the stated inequality.

5. QUANTITATIVE CONTINUATION

We now propagate the conjugate-node bound of Lemma 4.3 to the complex region required by the test (Section 2.4).

Lemma 5.1 (Uniform improvement of the pure kernel). *With the dyadic parameters of Theorem 7.1, factor the finite PSD matrix P_+ as a Gram. Let $V(\xi, \eta)$ be its holomorphic vector of wedge evaluations, chosen so that $\|V(\xi, \eta)\|^2 = P_+[\xi, \eta]$, and put*

$$V_\epsilon(\alpha, \beta) = V(i\epsilon^{-\epsilon(\alpha+i\beta)}, -i\epsilon^{-\epsilon(\alpha-i\beta)}).$$

Then, throughout the fixed region

$$1 \leq \Re \alpha \leq 2, \quad |\Im \alpha| \leq 256, \quad |\Re \beta| \leq 256, \quad |\Im \beta| \leq \frac{1}{2},$$

one has

$$\epsilon^4 \|V_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{2^{-17}}. \quad (5.1)$$

Proof. Boundary bounds. The vector is entire, since the finite feature vector is polynomial before composition. More explicitly, a real factorization $P_+ = \Gamma^T \Gamma$ gives $V = \Gamma f$, so the stated squared-norm identity holds also at complex arguments. The two nodes have moduli

$$|\xi| = e^{-\epsilon(\Re \alpha - \Im \beta)}, \quad |\eta| = e^{-\epsilon(\Re \alpha + \Im \beta)}.$$

If $\Re \alpha - |\Im \beta| \geq 1/4$, (4.3) and $1 - e^{-\epsilon/2} \geq \epsilon/4$ give the global bound

$$\|V_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{-4}. \quad (5.2)$$

Here 2^{11} would suffice; the larger constant will be common to both boundary estimates. On the real rectangle $1/2 \leq \alpha \leq 1024 + 1/2$, $|\beta| \leq 512$, the nodes are conjugates, with $\rho = e^{-2\epsilon\alpha}$ and $\vartheta = \pi/2 - \epsilon\beta$. As $\epsilon \leq 2^{-16} < 2^{-12}$, they satisfy $\rho \geq 1/2$, $|\sin \vartheta| \geq 1/2$, and $1 - \rho \geq \alpha\epsilon \geq \epsilon/2$. The infinite repeated-node expression satisfies

$$Q_{0,\infty}[\xi, \bar{\xi}] \leq \frac{1}{2\rho \sin^2 \vartheta (1 - \rho)^2} \leq 16\epsilon^{-2}.$$

Lemma 4.3 and (4.10) add at most ϵ^{-2} , giving

$$\|V_\epsilon(\alpha, \beta)\|^2 \leq 17\epsilon^{-2} \leq 2^{20} \epsilon^{-2}. \quad (5.3)$$

The comparison principle. Fix one of the variables α, β and let the other range over a closed rectangle. Suppose $\|V_\epsilon\|^2 \leq M_*$ on the boundary and $\|V_\epsilon\|^2 \leq M_* \epsilon^\theta$ on one distinguished side. If h is harmonic, zero on the other sides, and between zero and one on the distinguished side, then

$$\|V_\epsilon\|^2 \leq M_* \epsilon^{\theta h} \quad \text{on the rectangle.} \quad (5.4)$$

Indeed $\log \|V_\epsilon\|^2 - \log M_* - \theta h \log \epsilon$ is subharmonic and at most zero on the boundary, since $\log \epsilon < 0$. The maximum principle proves the claim; zeros of V_ϵ give $-\infty$ and cause no problem. The argument does not depend on the dimension of the values of V_ϵ .

First rectangle. Fix real $\beta \in [-512, 512]$. In the upper α -rectangle $1/2 \leq \Re \alpha \leq 1024 + 1/2$, $0 \leq \Im \alpha \leq 512$, use

$$h_1(\alpha) = \sin \frac{\pi(\Re \alpha - 1/2)}{1024} \frac{\sinh(\pi(512 - \Im \alpha)/1024)}{\sinh(\pi/2)}. \quad (5.5)$$

It is harmonic, zero on the sides and top, and between zero and one on the bottom. On $1 \leq \Re \alpha \leq 2$, $0 \leq \Im \alpha \leq 256$, sine concavity gives a factor at least $1/1024$, while $\sinh(\pi/4)/\sinh(\pi/2) = 1/(2 \cosh(\pi/4)) > 1/4$. Thus $h_1 > 2^{-12} > \theta_1 := 2^{-13}$. Apply (5.4) to (5.2)–(5.3), and reflect the rectangle to cover negative imaginary parts. The result is

$$\|V_\epsilon(\alpha, \beta)\|^2 \leq 2^{20} \epsilon^{-4+2\theta_1}, \quad 1 \leq \Re \alpha \leq 2, |\Im \alpha| \leq 256, \beta \in [-512, 512]. \quad (5.6)$$

Second rectangle. Fix such a complex α . In the β -rectangle $-512 \leq \Re \beta \leq 512$, $0 \leq \Im \beta \leq 3/4$, (5.2) remains valid because $\Re \alpha - |\Im \beta| \geq 1/4$. Its bottom has (5.6). Use

$$h_2(\beta) = \cos \frac{\pi \Re \beta}{1024} \frac{\sinh(\pi(3/4 - \Im \beta)/1024)}{\sinh(3\pi/4096)}. \quad (5.7)$$

On $|\Re \beta| \leq 256$, $0 \leq \Im \beta \leq 1/2$, its cosine factor exceeds $1/2$. The other factor is at least

$$\frac{\sinh(\pi/4096)}{\sinh(3\pi/4096)} = \frac{1}{3 + 4 \sinh^2(\pi/4096)} > \frac{1}{4}.$$

Consequently $h_2 > 1/8 > \theta_2 := 2^{-5}$. A second comparison and its reflection give (5.1), because $2\theta_1\theta_2 = 2^{-17}$. $\square$

Both rectangles stay strictly inside the domain allowed by (4.3).

6. AN EXACT NEGATIVE TANGENT DIRECTION

For speed pairs with positive real parts define the Hermitian kernel

$$S_F((a, b), (a', b')) = 2 \left\{ \frac{2}{A_0^2} + \frac{1}{C_0^2} - \frac{1}{A_0 C_0} \right\}, \quad \begin{aligned} A_0 &= (a + \bar{a}')(b + \bar{b}'), \\ C_0 &= (a + b)(\bar{a}' + \bar{b}'). \end{aligned} \quad (6.1)$$

To see the limit directly, substitute

$$\xi = ie^{-\epsilon a}, \quad \eta = -ie^{-\epsilon b}, \quad \zeta = \overline{ie^{-\epsilon a'}}, \quad \omega = \overline{-ie^{-\epsilon b'}}.$$

Then $A/\epsilon^2 \rightarrow A_0$, $C/\epsilon^2 \rightarrow C_0$, and $\Delta \rightarrow 4$. Consequently $\epsilon^4 L_\infty \rightarrow S_F$ in (4.2): S_F is the tangent kernel of the corner. Section 7 makes this limit quantitative at the test points.

Lemma 6.1 (Exact negative pairing). *On the points and coefficients (1.2),*

$$c^* S_{FC} < -2^{42}. \quad (6.2)$$

Proof. For $\ell = (r, j, \sigma)$ and $h = (r', j', \sigma')$, the entry denominators are positive integers:

$$A_0 = (r + r')^2 + (\sigma j - \sigma' j')^2, \quad C_0 = 4rr'.$$

Exact integer summation gives the rational enclosure

$$\begin{aligned} c^* S_{FC} &\geq -111807741093765987869670624504700 \cdot 2^{-64}, \\ c^* S_{FC} &\leq -111807741093765987869670623456124 \cdot 2^{-64} < -2^{42}. \end{aligned} \quad (6.3)$$

To obtain it, round each of the 1024^2 weighted entries downward at denominator 2^{64} ; the upper endpoint adds one unit per entry, so the numerators differ by exactly 1024^2 . Appendix B specifies the integer algorithm completely and gives an independent 513-term regrouping of the same sum. $\square$

The energy identity (2.3) explains each term of S_F . For a single exponential $e_\ell(s, t) = e^{-a_\ell s - b_\ell t}$ one has $\mathcal{H}e_\ell = e_\ell / (a_\ell + b_\ell)$, and with A_0, C_0 as in (6.1) for the pair (ℓ, h) ,

$$\int_{\mathbb{R}_+^2} st e_\ell \bar{e}_h ds dt = \frac{1}{A_0^2}, \quad \int_{\mathbb{R}_+^2} \mathcal{H}e_\ell \overline{\mathcal{H}e_h} ds dt = \frac{1}{A_0 C_0}, \quad \frac{1}{(a_\ell + b_\ell)^2 (\bar{a}_h + \bar{b}_h)^2} = \frac{1}{C_0^2}. \quad (6.4)$$

Summing against $c_\ell c_h$, with the real-coefficient convention of Section 2.3, gives (2.3) term by term. The signed pairs in (1.2) make ϕ antisymmetric and cancel the middle term. Their exponential difference is $-2ie^{-r(s+t)} \sin(j(s-t))$; its squared factor explains the four in (B.3). The negative channel in (2.3) is a diagonal-tail Hardy energy. Removing it makes the exact pairing positive.

7. THE FINITE-SCALE CONTRADICTION

All objects of the quantitative obstruction outlined in Section 1 are now defined: the averaged Gram matrix and its positive semidefinite blocks M, P_+, P_- (Lemma 3.2), their four-slot generating kernels (the kernel conventions of Section 3), and the points and coefficients (1.2). We state it precisely and prove it.

Theorem 7.1 (Uniform finite-scale obstruction). *Let $k \geq 16$ be an integer, $\epsilon = 2^{-k}$, and $m = \epsilon^{-2} = 2^{2k}$. If $\mathcal{B}_m$ is a sum of squares, then every averaged positive semidefinite Gram matrix in Lemma 3.2 gives a positive semidefinite Hermitian matrix*

$$\mathbf{G}_m = \left[\epsilon^4 \left(M + \frac{P_+ + P_-}{2} \right) (X_\ell, \overline{X_h}) \right]_{\ell, h \in I}, \quad X_\ell = (ie^{-\epsilon a_\ell}, -ie^{-\epsilon b_\ell}), \quad (7.1)$$

and its nonnegative quadratic pairing $\Phi_m = c^* \mathbf{G}_m c$ satisfies

$$\Phi_m < -2^{42} + 2^{90} m^{-1/2} + 2^{80} m^{-1/262144}. \quad (7.2)$$

All constants are independent of the sum-of-squares decomposition, the Gram corrections, and the rank of a Gram factor.

Here M, P_+, P_- are the three positive semidefinite blocks of the averaged Gram matrix (Lemma 3.2), viewed as four-slot kernels and evaluated at the nodes X_ℓ . The upper bound is negative once m is large enough; Corollary 1.2 makes an explicit choice of k .

Proof. Fix $k \geq 16$ and a positive semidefinite Gram matrix of the corner, averaged as in Lemma 3.2. Write $\mathbf{L}_m$ and $\mathbf{R}_m$ for the matrices of evaluated entries of L_m and $\mathcal{R}P_+$, respectively, on this fixed list of points. Write $X_\ell = (\xi_\ell, \eta_\ell)$ for the nodes (7.1); since $b_\ell = \bar{a}_\ell$, $\eta_\ell = \bar{\xi}_\ell$. The matrix $\mathbf{G}_m$ is PSD: each summand is an evaluated PSD kernel on the same finite list. Its pure and mixed features may live in different spaces; their direct sum is still a Gram. Thus $\Phi_m \geq 0$.

The crucial realigned entry is exactly a Hermitian diagonal:

$$(\mathcal{R}P_+)(X_\ell, \overline{X_h}) = P_+(\xi_\ell, \bar{\xi}_h, \bar{\xi}_\ell, \xi_h) = P_+[\xi_\ell, \bar{\xi}_h]. \quad (7.3)$$

The array of such entries need not be PSD. If $\ell = (r, j, \sigma)$ and $h = (r', j', \sigma')$, its coordinates for Lemma 5.1 are

$$\alpha = (r + r')/2 + i(\sigma j - \sigma' j')/2, \quad \beta = (\sigma j + \sigma' j')/2 - i(r - r')/2.$$

Indeed $\alpha + i\beta = a_\ell$ and $\alpha - i\beta = \bar{a}_h$, which gives exactly the arguments $\xi_\ell, \bar{\xi}_h$ in the pure kernel. They satisfy every bound in that lemma. Therefore

$$\left| \epsilon^4 c^* \mathbf{R}_m c \right| \leq \|c\|_1^2 2^{20} \epsilon^{2^{-17}} < 2^{80} \epsilon^{2^{-17}}. \quad (7.4)$$

It remains to estimate the finite baseline. At the test points each factor of A and of C has the form $1 - e^{-\epsilon\chi}$, where χ is a sum of two of the speeds $a_\ell, b_\ell, \bar{a}_h, \bar{b}_h$; thus $\Re\chi \geq 2$ and $|\chi| < 2^{10}$. Writing $(1 - e^{-\epsilon\chi})/(\epsilon\chi) = \int_0^1 e^{-\epsilon\chi u} du$ gives

$$\left| \frac{1 - e^{-\epsilon\chi}}{\epsilon\chi} - 1 \right| \leq \epsilon|\chi|/2 \leq 2^{10}\epsilon.$$

Hence $A/\epsilon^2 = A_0(1 + \delta_A)$ and $C/\epsilon^2 = C_0(1 + \delta_C)$, with $|\delta_A|, |\delta_C| \leq 2^{12}\epsilon \leq 1/16 < 1/4$. For $|\delta| \leq 1/4$, use $|(1 + \delta)^{-1} - 1| \leq 2|\delta|$ and $|(1 + \delta)^{-2} - 1| \leq 6|\delta|$. Since $A_0, C_0 \geq 4$ and $|\Delta| \geq 1$, substitution in (4.2) yields

$$|\epsilon^4 L_\infty(X_\ell, \bar{X}_h) - S_F(\ell, h)| \leq 2^{20}\epsilon.$$

For the bound on Δ , its factors are $1 + e^{-\epsilon\chi}$, and $|\Im(\epsilon\chi)| \leq 512\epsilon < 1$, so the exponential has positive real part. The estimate above follows by bounding the three reciprocal errors by $3 \cdot 2^{12}\epsilon$ in total and the remaining term by $4\epsilon^4$. Adding the finite tail from (4.10) gives, entrywise,

$$|\epsilon^4 L_m(X_\ell, \bar{X}_h) - S_F(\ell, h)| \leq 2^{30}\epsilon. \quad (7.5)$$

All slot moduli here are at most $e^{-\epsilon}$, so Lemma 4.3 applies.

Finally apply (3.9), (6.2), (7.4), (7.5), and $\|c\|_1^2 < 2^{60}$:

$$\begin{aligned} 0 \leq \Phi_m &= \epsilon^4 c^* \mathbf{L}_m c - \epsilon^4 c^* \mathbf{R}_m c \\ &< -2^{42} + 2^{90}\epsilon + 2^{80}\epsilon^{2^{-17}} \\ &= -2^{42} + 2^{90}m^{-1/2} + 2^{80}m^{-1/262144}. \end{aligned} \quad (7.6)$$

This proves Theorem 7.1. □

Theorem 7.1 implies Corollary 1.2, and hence the negative assertion of Theorem 1.1.

8. THE FINITE POSITIVE RANGE

The positive results concern the literal polynomial (1.1). They can be verified by finite rational arithmetic, without using an eigenvalue computed in floating point.

Proposition 8.1 (Exact positive certificates). *The polynomial F_N is a sum of squares for every $1 \leq N \leq 50$. For $2 \leq N \leq 20$, this assertion is also formally verified in Lean 4 with Mathlib.*

Proof. The first two cases are

$$F_1 = 0, \quad F_2 = 4z_{-1,0}^2 + 4z_{0,1}^2.$$

For the remaining orders the accompanying exact certificates give a chain of 48 polynomial identities, one for each step $n = 2, \dots, 49$:

$$F_{n+1} = a_n F_n \circ \psi_n + z^T Q_n z, \quad a_n > 0, \quad Q_n \succeq 0. \quad (8.1)$$

Here a_n is rational, ψ_n is a rational linear substitution of the symbol variables, and Q_n is a rational Gram matrix on the wedges of F_{n+1} , central squares included. The early steps use scalar or identity substitutions, the later ones geometric damping of the signed symbols. The accompanying checker reconstructs each F_n from the matrix definition, expands every identity (8.1) exactly, and verifies the forced kernel and the positive definiteness of the compressed rational blocks; the certificate formats are specified in the accompanying certificate documentation and verification map. A positive scalar multiple, a linear substitution and a sum all preserve sums of squares, so induction proves the statement through $N = 50$.

For completeness, the last identity is

$$F_{50} = F_{49} \circ \psi_{49} + z^T Q_{49} z, \quad \psi_{49}(w_\lambda) = (97/98)^{|\lambda|} w_\lambda. \quad (8.2)$$

Use the noncentral wedges z_{ab} , $a < b$, with $a, b \in \{-49, \dots, -1, 1, \dots, 49\}$. In each mixed gap $g = b - a = 50, \dots, 98$, remove the first coordinate in lexicographic wedge order, and denote the retained principal submatrix of Q_{49} by $\widehat{Q}_{49}$. The checker prints the exact lower bound

$$\widehat{Q}_{49} \succeq 0.015624875921 I \succ \frac{1}{65} I,$$

a rational bound rounded down from an exact residual estimate. After adjoining the forced zero directions, the retained matrix is congruent to the full noncentral Gram matrix; the central squares are checked separately. This establishes positivity exactly. The formal assertion through order 20 is the theorem stated in Appendix E; it proves the same polynomial and the same real homogeneous sum-of-squares predicate. $\square$

8.1. Rational certificates. A rational Gram matrix $Q \succeq 0$ with $F_N = z^T Q z$ proves that F_N is a sum of squares in the unrestricted real sense: a real factorization of Q writes $z^T Q z$ as a sum of squares of real linear combinations of the wedges, whose coefficients need not be rational. Conversely, by Lemma 3.1 every real sum of squares arises in this way from a real Gram matrix. Because every positive semidefinite Gram matrix of F_N has forced kernel vectors, no bound $Q \succeq \delta I$ with $\delta > 0$ holds on the whole wedge space; a positive lower bound, such as the one above, refers to its stated principal block. Bounds relative to the pure diagonal are the subject of the next section.

9. THE EXCHANGE CRITERION AND CRITICALITY

The same wedge identities describe the sum-of-squares problem exactly at each fixed order, and show what the margins of positive certificates can be.

Fix $n \geq 3$ and use one-based depths $1 \leq p < n$:

$$w_p^- = w_{p-n}, \quad w_p^+ = w_{n-p}, \quad u_{pq} = w_p^- \wedge w_q^-, \quad v_{pq} = w_p^+ \wedge w_q^+, \quad z_{pq}^{\text{mix}} = w_p^- \wedge w_q^+.$$

Both pure sides are oriented by increasing depth, so v_{pq} is the negative of the wedge ordered by increasing label. Unlike the stabilized corner, these variables include all noncentral diagonals.

The central symbol splits off as explicit positive squares:

$$F_n = F_n|_{w_0=0} + \sum_{\ell \neq 0} 2n(n - |\ell|) z_{0\ell}^2. \quad (9.1)$$

Indeed the central shift is the identity, commutes with every shift, and is Frobenius orthogonal to the other shifts. Therefore F_n is SOS if and only if its noncentral restriction is SOS.

9.1. **The pure budget.** On $W_n = \Lambda^2 \mathbb{R}^{n-1}$ define

$$\begin{aligned} D_n &= \text{diag}(2pq)_{1 \leq p < q < n}, \\ (J_n)_{pq,rs} &= \mathbf{1}_{q-p=s-r} \min(p, r, n-q, n-s), \quad p < q, r < s. \end{aligned} \quad (9.2)$$

Let $\mathcal{P}_{n-1}$ be the pure Plücker span. On mixed coordinates put

$$(M_n^0)_{pq,rs} = 2pq \mathbf{1}_{(p,q)=(r,s)} - 2 \mathbf{1}_{p+q=r+s} \min(p, q, r, s, n-p, n-q, n-r, n-s). \quad (9.3)$$

For a symmetric matrix Ξ on W_n , the realignment $\mathcal{R}(\Xi)$ is the mixed matrix (3.3). Let

$$\begin{aligned} \mathcal{L}_n &= \{\Xi = \Xi^T : \mathcal{R}(\Xi) s_g = 0, 2 \leq g \leq n\}, \\ \mathcal{C}_n &= \{\Xi \in \mathcal{L}_n : M_n^0 + \mathcal{R}(\Xi) \succeq 0\}, \quad s_g = \sum_{p+q=g} \mathbf{e}_{pq}. \end{aligned} \quad (9.4)$$

Define the pure-completion norm by

$$\nu_n(Y) = \max_{\substack{Z \succeq 0, Z \perp \mathcal{P}_{n-1} \\ \text{tr}(D_n Z) = 1}} \text{tr} |Z^{1/2} Y Z^{1/2}|, \quad (9.5)$$

where $|Y|$ is the spectral absolute value, and define the budget

$$\gamma_n = \min_{\Xi \in \mathcal{C}_n} \nu_n(2J_n + \Xi). \quad (9.6)$$

The normalizations in this section differ from the zero-based corner only by the stated index shift and by naming the positive minimum kernel J_n , so that the pure cross block is $-(2J_n + \Xi)$.

Theorem 9.1 (Exact exchange criterion). *For every $n \geq 3$, the maximum in (9.5) and the minimum in (9.6) are attained, ν_n is a norm, and*

$$F_n \text{ is SOS} \iff \gamma_n \leq 1.$$

Equivalently, the latter condition is the existence of $\Xi \in \mathcal{C}_n$ and $E \in \mathcal{P}_{n-1}$ such that

$$D_n + E \pm (2J_n + \Xi) \succeq 0. \quad (9.7)$$

If $\gamma_n < 1$, such a certificate can be chosen rational, with both pure blocks positive definite and the mixed block positive definite on the complement of the forced means.

Proof. The complete wedge-frame and symmetry argument of Lemma 3.2 applies to the noncentral restriction. For the signed shifts S_{p-n} and S_{n-q} , multiplication and cancellation of the overlapping strings give

$$\begin{aligned} &\langle [S_{p-n}, S_{n-q}], [S_{r-n}, S_{n-s}] \rangle_F \\ &= 2 \mathbf{1}_{p-q=r-s} \min(p, q, r, s, n-p, n-q, n-r, n-s). \end{aligned}$$

Indeed different offsets are orthogonal; on a common offset the two remaining end strings have this common length. The norm terms have weights $2pq$. Applying (3.8) to each raw mixed pair with equal offset sends it to a mixed pair with equal total and a pure pair with equal gap. This gives M_n^0 and J_n above, and the averaged Gram is

$$\begin{pmatrix} D_n + E & -(2J_n + \Xi) & 0 \\ -(2J_n + \Xi) & D_n + E & 0 \\ 0 & 0 & M_n^0 + \mathcal{R}(\Xi) \end{pmatrix}.$$

The cross sign follows from $u_{pq}v_{rs} - z_{pr}^{\text{mix}}z_{qs}^{\text{mix}} + z_{ps}^{\text{mix}}z_{qr}^{\text{mix}} = 0$: specifying the parallel mixed entry to be $\Xi_{pq,rs}$ forces the pure cross entry $-2(J_n)_{pq,rs} - \Xi_{pq,rs}$. All pure four-forms remain free. The sum and difference of the pure sides give (9.7).

For totals $g \leq n$, the row-sum identity

$$\sum_{r=1}^{g-1} \min(p, g-p, r, g-r) = p(g-p)$$

makes M_n^0 a weighted complete-graph Laplacian. For $g > n$, reversal $p \mapsto n-p$, $q \mapsto n-q$ gives the block at $2n-g$ plus $2n(g-n)I$. Hence $M_n^0 \succeq 0$ and its kernel is exactly the span of $s_2, \dots, s_n$. The least-coefficient argument of Lemma 4.1 forces these same means for every positive Gram. Thus imposing $\mathcal{L}_n$ loses no feasible matrix.

The set $\mathcal{C}_n$ contains 0 and is closed and convex. Its mixed diagonal is fixed. Positive semidefiniteness bounds every entry by the geometric mean of its diagonal entries, and the parallel entries recover every entry of Ξ . This proves compactness.

For a fixed symmetric Y , consider

$$\inf\{\tau : \exists E \in \mathcal{P}_{n-1}, \quad \tau D_n + E \pm Y \succeq 0\}. \quad (9.8)$$

It is finite and nonnegative. In a bounded sublevel set, the diagonals of both positive matrices are bounded, since E has zero diagonal; positive semidefiniteness bounds their off-diagonal entries and hence E . Closedness gives attainment. Taking $E = 0$ and τ sufficiently large gives strict feasibility. Finite-dimensional semidefinite duality identifies (9.8) with

$$\max\{\langle Y, Z_+ - Z_- \rangle : Z_{\pm} \succeq 0, Z_+ + Z_- \perp \mathcal{P}_{n-1}, \langle D_n, Z_+ + Z_- \rangle = 1\}.$$

This dual set is nonempty and compact. For fixed $Z = Z_+ + Z_-$, the maximum is $\text{tr}|Z^{1/2}YZ^{1/2}|$: on the range of Z , the difference $Z_+ - Z_-$ is $Z^{1/2}UZ^{1/2}$ with $-I \preceq U \preceq I$. The argument includes singular Z . It proves (9.5) and its attainment. The formula is homogeneous and subadditive; it separates nonzero Y because a positive multiple of the identity is an admissible Z . Thus it is a norm.

Continuity and compactness give attainment in (9.6). An attained value $\tau \leq 1$ in (9.8) can be increased to 1 by adding $(1-\tau)D_n \succeq 0$. This proves the equivalence, including equality at $\gamma_n = 1$.

Finally, if $\gamma_n < 1$, move Ξ a sufficiently small distance toward 0. The mixed block becomes positive definite on the forced kernel complement and the pure budget remains below 1. Rational points are dense in the rational subspaces $\mathcal{L}_n$ and $\mathcal{P}_{n-1}$. Approximate within the resulting strict inequalities. This gives the rational certificate claimed. $\square$

Theorem 9.2 (Criticality of the pure budget). *For every $n \geq 3$, $\frac{1}{2} \leq \gamma_n \leq 4$. For every $n \geq 128$,*

$$\gamma_n \geq 1 - \frac{33[\log_2 n]^2}{n^2}. \quad (9.9)$$

If an SOS certificate has pure blocks $D_n + E \pm (2J_n + \Xi) \succeq \mu D_n$, then $\mu \leq 1 - \gamma_n$. In particular its relative pure margin is at most the displayed deficit when $n \geq 128$.

Proof. Appendix D proves the quantitative bounds from an explicit decomposable wedge test and the complete low means. For the margin assertion, subtract μD_n from both pure blocks. The same E, Ξ is then feasible at budget $1 - \mu$, so $\gamma_n \leq 1 - \mu$. $\square$

9.2. Consequences for certificates. Any feasible certificates at increasing orders must therefore lose their relative pure margin at least at the rate imposed by (9.9). This is the precise meaning of criticality here. The theorem gives a lower bound for γ_n , not convergence to 1. Indeed Theorem 1.1 and the exact criterion imply $\gamma_n > 1$ throughout the large-order non-SOS range.

10. BOUNDARY LAYERS AND INCREMENTS

A natural route to sums of squares at every order is induction on the order. Define the increment in a common signed-symbol ring by

$$R_n = (n-1)F_{n+1} - (n+1)F_n, \quad n \geq 2, \quad (10.1)$$

so that

$$F_{n+1} = \frac{n+1}{n-1}F_n + \frac{1}{n-1}R_n.$$

If every increment were SOS, every F_n would be SOS. This section shows that the increments eventually fail to be SOS. The reason is that the stabilized corner has a simple discrete derivative, the boundary-layer polynomial; the relation is an algebraic consequence of depth weights and is independent of the proof of Theorem 7.1.

In this section use one-based depths $1 \leq p, q \leq \kappa$ and the same pure and mixed wedge orientations as before. Put $d_{pq} = p + q - 1$ and

$$\mathcal{O}_\delta = \sum_{p+q=\delta} z_{pq}^{\text{mix}}, \quad s_g = \sum_{p+q=g} z_{pq}^{\text{mix}}, \quad U_h = \sum_{p=1}^{\kappa-h} u_{p,p+h}, \quad V_h = \sum_{p=1}^{\kappa-h} v_{p,p+h}.$$

The boundary-layer polynomial is

$$\text{BL}_\kappa = 2 \sum_{p,q=1}^{\kappa} d_{pq} (z_{pq}^{\text{mix}})^2 + 2 \sum_{p < q} d_{pq} (u_{pq}^2 + v_{pq}^2) - 2 \sum_{\delta} \mathcal{O}_\delta^2. \quad (10.2)$$

Its weights are linear in depth. The weights of $\mathcal{B}_\kappa$ are products pq ; these are different polynomials.

Applying the Plücker relation to pairs of mixed wedges of the same offset gives the exchange identity

$$\sum_{\delta} \mathcal{O}_\delta^2 = \sum_g s_g^2 + 2 \sum_h U_h V_h. \quad (10.3)$$

Each mixed antidiagonal pair occurs once, and each unordered pair of pure wedges twice. This is the linear-weight counterpart of the exchange that gives the matrices in Lemma 3.2.

Set $\mathcal{B}_0 = 0$. Let π_j delete depths $1, \dots, j-1$ and relabel the remaining ones starting at 1. Thus $\text{BL}_{\kappa-j+1} \circ \pi_j$ uses only the tail of the original variables. Direct summation of depth weights gives

$$\mathcal{B}_\kappa = \sum_{j=1}^{\kappa} \text{BL}_{\kappa-j+1} \circ \pi_j, \quad \mathcal{B}_\kappa^{\text{tail}} := \sum_{j=2}^{\kappa} \text{BL}_{\kappa-j+1} \circ \pi_j = \mathcal{B}_{\kappa-1} \circ \pi_2. \quad (10.4)$$

For example, the diagonal identity is $\sum_{j=1}^{\min(p,q)} (p+q-2j+1) = pq$. The offset kernel sums its common-depth shifts to $\min(p, q, r, s)$. These prove every coefficient in (10.4); also $\mathcal{B}_\kappa = \mathcal{B}_\kappa^{\text{tail}} + \text{BL}_\kappa$.

For the increment (10.1), the variables at offsets $\pm n$ have zero weight in F_n . For this step, retain the outer κ depths $\pm(n+1-p)$, $1 \leq p \leq \kappa$, and call this restriction $\iota_{n,\kappa}$. When $n \geq \max(2, 2\kappa-1)$, direct shift multiplication gives

$$F_{n+1} \circ \iota_{n,\kappa} = \mathcal{B}_\kappa, \quad F_n \circ \iota_{n,\kappa} = \mathcal{B}_\kappa^{\text{tail}}, \quad R_n \circ \iota_{n,\kappa} = (n-1)\text{BL}_\kappa - 2\mathcal{B}_\kappa^{\text{tail}}. \quad (10.5)$$

The onset ensures that the two opposite boundary strings are disjoint. The old weights in the restricted F_n are $(p-1)(q-1)$ and the minimum is $\min(p, q, r, s) - 1$, exactly as the tail formula requires.

Corollary 10.1 (Failure of the all-order increment condition). *There is a least integer κ_* for which BL_{κ_*} is not SOS. For every $\kappa \geq \kappa_*$, BL_κ is not SOS, and for every $n \geq \max(2, 2\kappa_* - 1)$, R_n is not SOS.*

Proof. If every BL_κ were SOS, (10.4) would make every $\mathcal{B}_\kappa$ SOS, contradicting Corollary 1.2. Restriction to the first κ_* depths sends BL_κ to BL_{κ_*} , which proves the all-larger-width assertion. By minimality, every summand in $\mathcal{B}_{\kappa_*}^{\text{tail}}$ is SOS. If R_n were SOS at one of the stated orders, (10.5) would give

$$(n-1)\text{BL}_{\kappa_*} = R_n \circ \iota_{n,\kappa_*} + 2\mathcal{B}_{\kappa_*}^{\text{tail}}$$

as a sum of squares, a contradiction. $\square$

11. THE INTERMEDIATE RANGE

The answer is positive for every $N \leq 50$ and negative for every $N \geq 2^{9961475}$. By Theorem 9.1, the remaining orders are decided by the sign of $\gamma_N - 1$, which the present estimates do not determine. Theorem 9.2 forces the relative pure margins of any certificates toward zero as the order grows, and Theorem 1.1 gives $\gamma_N > 1$ for $N \geq 2^{9961475}$; neither result locates a transition.

Remark 11.1 (An expectation at moderate orders). The exact positive certificates through order 50 and the small normalized defect $2^{-18} < \delta_{\text{test}} < 2^{-16}$ of the tangent witness (Appendix B) suggest that the positive SOS range may extend to orders in the hundreds. This is an expectation without a specified endpoint, not a proved extension of the positive range. The continuation estimate explains why the present obstruction becomes effective only much later; it cannot exclude a different obstruction at a smaller order.

11.1. Open questions. Is there an integer N_* such that F_N is a sum of squares exactly for the orders $1 \leq N < N_*$? If so, $51 \leq N_* \leq 2^{9961475}$. A first failure exists by Theorem 1.1; the question is whether the sum-of-squares property can return at a larger order. No monotonicity theorem is known for γ_N or for the forms F_N .

Further questions concern the first non-SOS order, improvements of the sufficient threshold through a different tangent test or a stronger continuation estimate, and exact positive certificates at intermediate orders.

APPENDIX A. THE GENERATING KERNELS

We derive the generating kernels (4.2), in the notation of Section 4. Their order of singularity at the boundary of the polydisk is responsible for the tangent scale. All four

variables lie in the open unit disk, so the series and their rearrangements converge absolutely. For the pure diagonal,

$$D_\infty = 2 \sum_{p < q} (p+1)(q+1)(\xi^p \eta^q - \xi^q \eta^p)(\zeta^p \omega^q - \zeta^q \omega^p).$$

Expanding, pairing the two orders p, q , and cancelling the equal-index terms gives $2(A^{-2} - \Delta^{-2})$.

For the mixed minimum, use

$$\min(p+1, q+1, r+1, s+1) = \sum_{h \geq 0} \mathbf{1}_{h \leq p, q, r, s}.$$

Shifting all four indices by h preserves the equation $p+q = r+s$ and contributes Π^h . The unweighted total kernel is

$$\sum_{g \geq 0} \left(\sum_{p=0}^g \xi^p \eta^{g-p} \right) \left(\sum_{r=0}^g \zeta^r \omega^{g-r} \right) = \frac{1 - \Pi}{A\Delta}.$$

For distinct ξ, η and ζ, ω , this follows by using $\sum_{p=0}^g \xi^p \eta^{g-p} = (\xi^{g+1} - \eta^{g+1})/(\xi - \eta)$ and summing four geometric series. Equality at repeated variables follows by continuity. Summing the common shifts divides it by $1 - \Pi$ and gives $1/(A\Delta)$, proving the formula for B_∞ .

For the gap kernel write $q = p + h, s = r + h, h \geq 1$. Its wedge factors are $(\xi\eta)^p(\eta^h - \xi^h)$ and $(\zeta\omega)^r(\omega^h - \zeta^h)$. The starting-depth sum is

$$\sum_{p, r \geq 0} \min(p+1, r+1)(\xi\eta)^p(\zeta\omega)^r = \frac{1}{C(1 - \Pi)}.$$

The remaining gap sum is

$$\sum_{h \geq 1} (\eta^h - \xi^h)(\omega^h - \zeta^h) = \frac{\eta\omega}{1 - \eta\omega} - \frac{\eta\zeta}{1 - \eta\zeta} - \frac{\xi\omega}{1 - \xi\omega} + \frac{\xi\zeta}{1 - \xi\zeta} = \frac{(1 - \Pi)(\Delta - A)}{A\Delta}.$$

Multiplication by the cross coefficient -2 gives $K_\infty = -2(\Delta - A)/(A\Delta C)$ and hence $Q_{0,\infty} = D_\infty + K_\infty$. Finally, realignment interchanges A with C and leaves Δ fixed. In fact the realigned term is

$$\mathcal{R}Q_{0,\infty} = 2(C^{-2} - \Delta^{-2}) - 2(\Delta - C)/(A\Delta C).$$

Adding $B_\infty + D_\infty$ leaves

$$2 \left(\frac{2}{A^2} + \frac{1}{C^2} - \frac{2}{\Delta^2} - \frac{1}{A\Delta} - \frac{\Delta - C}{A\Delta C} \right).$$

The last two fractions combine to $-1/(AC)$, giving the last line of (4.2).

APPENDIX B. THE EXACT INTEGER WITNESS

This appendix specifies the rational sign check of Lemma 6.1 completely. No approximation of a transcendental number enters the check: for $\ell = (r, j, \sigma)$ and $h = (r', j', \sigma')$ the complex speeds only lead to the positive integers

$$A_0 = (r + r')^2 + (\sigma j - \sigma' j')^2, \quad C_0 = 4rr'.$$

On these points,

$$(S_F)_{\ell h} = \frac{4C_0^2 + 2A_0^2 - 2A_0C_0}{A_0^2C_0^2}. \quad (\text{B.1})$$

The integer weights $\hat{c}_j$ are given by the floor formula (1.2). Over the 1024^2 ordered pairs compute

$$J = \sum_{\ell, h} \left\lfloor \frac{2^{64} c_\ell c_h (4C_0^2 + 2A_0^2 - 2A_0C_0)}{A_0^2C_0^2} \right\rfloor, \quad (\text{B.2})$$

with every quotient rounded toward negative infinity. Each summand bounds its exact rational entry from below with error less than one unit at denominator 2^{64} . Consequently

$$\frac{J}{2^{64}} \leq c^* S_{FC} \leq \frac{J + 1048576}{2^{64}}.$$

The exact outputs are

$$\begin{aligned} J &= -111807741093765987869670624504700, \\ J + 1048576 &= -111807741093765987869670623456124, \\ \|c\|_1 &= 859652168. \end{aligned}$$

The first two are the numerators of the endpoints in (6.3). The upper numerator is smaller than -2^{106} , proving $c^* S_{FC} < -2^{42}$. All loops have fixed finite bounds; the formulas above are a full specification of the calculation.

B.1. A regrouped sum. Grouping the four sign choices gives a shorter exact summation. Define

$$d_l = \sum_{j, j'=1}^{256} \hat{c}_j \hat{c}_{j'} (\mathbf{1}_{|j-j'|=l} - \mathbf{1}_{j+j'=l}), \quad 0 \leq l \leq 512,$$

and

$$\mathcal{K}(l) = \sum_{r, r' \in \{1, 2\}} \left(\frac{2}{((r+r')^2 + l^2)^2} - \frac{1}{4rr'((r+r')^2 + l^2)} \right).$$

Expanding the four sign choices in (B.1) cancels the rank-one term, because $\sum_{\sigma, \sigma'} \sigma \sigma' = 0$, and gives

$$c^* S_{FC} = 4 \sum_{l=0}^{512} d_l \mathcal{K}(l). \quad (\text{B.3})$$

This is a 513-term exact rational sum after integer convolution of the weights, and it agrees with the full-entry enclosure. The factor four comes from replacing a difference of two exponentials by a sine.

B.2. Normalization. If the negative contribution $-2/(A_0C_0)$ is removed from every entry, the pairing becomes strictly positive; the accompanying checker verifies this, the enclosure above, and that the convolution (B.3) lies in it, all in exact arithmetic. Relative to the coefficient mass, the defect $\delta_{\text{test}} = -c^* S_{FC} / \|c\|_1^2$ satisfies

$$2^{-18} < \delta_{\text{test}} < 2^{-16}$$

by the same endpoints. This interval describes this particular test, not the smallest eigenvalue of S_F or an optimal sum-of-squares budget.

APPENDIX C. CONSTANTS AND UNIFORMITY

Every estimate in the negative proof is uniform over the full affine space of permitted Gram corrections after positivity has been imposed. The following table records each bound and the only inputs it depends on. The dimension and rank of a Gram factor are never among them, and no exponential is evaluated at an astronomical argument: the tail bounds (4.10) need only $2^{k-2} \geq 8k + 40$ for $k \geq 16$.

Estimate	Bound and dependence
Fixed test	1024 points, $r \leq 2$, $j \leq 256$; $\ c\ _1 < 2^{30}$ and $c^* S_{FC} < -2^{42}$. Only the integer formula (1.2).
Fixed coefficients	$16(p+1)(q+1)(r+1)(s+1)$ for the expanded coefficients of $Q_{0,\infty}$ and L_∞ . Only the explicit baseline matrices.
Mixed correction	$4\sqrt{(p+1)(q+1)(r+1)(s+1)}$. Only the fixed diagonal and positive semidefiniteness.
Truncation	$2^{30}\epsilon^{-8}e^{-\epsilon m/4}$. Only m, ϵ , with $0 < \epsilon \leq 1$.
Repeated-node correction	$2^{26}\epsilon^{-8}e^{-\epsilon m/2}$. The same parameters and the complete low means.
Combined tail	$\mathcal{E} = 2^{32}\epsilon^{-8}e^{-\epsilon m/4}$. For $m = \epsilon^{-2}$, $\epsilon = 2^{-k}$, $k \geq 16$: $\mathcal{E} \leq \epsilon^{-2}$ and $\epsilon^4 \mathcal{E} \leq \epsilon$.
Boundary bounds	$2^{20}\epsilon^{-4}$ globally and $2^{20}\epsilon^{-2}$ on the real rectangle. The latter uses the sharper $17\epsilon^{-2}$ estimate and $\epsilon \leq 2^{-12}$.
Rectangle weights	$\theta_1 = 2^{-13}$, $\theta_2 = 2^{-5}$; $2\theta_1\theta_2 = 2^{-17}$. Only the fixed rectangles.
Pure-kernel improvement	$2^{20}\epsilon^{2^{-17}}$ after multiplication by ϵ^4 . Only the preceding two bounds.
Baseline comparison	$2^{20}\epsilon$ before truncation, $2^{30}\epsilon$ after truncation. Only the fixed speeds, $\epsilon \leq 2^{-13}$, and $k \geq 16$.
Scalar losses	$2^{90}\epsilon$ and $2^{80}\epsilon^{2^{-17}}$. The entrywise bounds multiplied by $\ c\ _1^2 < 2^{60}$.

The passage from ϵ to m accounts for the exponent in Theorem 7.1:

$$\epsilon = m^{-1/2}, \quad \epsilon^{2^{-17}} = m^{-1/262144}.$$

At $k = 38 \cdot 2^{17} + 1 = 4980737$, the slow error equals $2^{42}2^{-2^{-17}} \leq 2^{42} - 2^{24}$, by the convexity inequality used in Corollary 1.2. The other error is less than one, so the budget is at most $-2^{24} + 1 < 0$. An entirely integer check of the slow-error bound is

$$(2^{18} - 1)^{2^{17}} > 2^{18 \cdot 2^{17} - 1}.$$

At the preceding integer $k = 38 \cdot 2^{17}$ the slow error is exactly 2^{42} and the budget is positive. Both errors decrease with k , so 4980737 is the least integer making this particular budget negative.

APPENDIX D. PROOF OF THE CRITICALITY ESTIMATE

We prove Theorem 9.2 in the notation of Section 9. A feasible budget τ satisfies

$$\tau D_n + E \pm (2J_n + \Xi) \succeq 0, \quad M = M_n^0 + \mathcal{R}(\Xi) \succeq 0, \quad \mathcal{R}(\Xi)s_g = 0 \quad (2 \leq g \leq n). \quad (\text{D.1})$$

Fix $0 < \rho < 1$. The pure vector

$$z_{pq} = \rho^{(p+q)/2} \sin \frac{\pi(q-p)}{2}$$

is decomposable: $z = f \wedge g$ for $f_p = \rho^{p/2} \cos(\pi p/2)$ and $g_p = \rho^{p/2} \sin(\pi p/2)$. Therefore $z^T E z = 0$ for every pure Plücker four-form. Put $d_n = z^T D_n z / 2$ and $c_n = z^T J_n z$. The scalar sums are

$$\begin{aligned} d_n &= \left(\sum_{\substack{1 \leq p < n \\ p \text{ odd}}} p \rho^p \right) \left(\sum_{\substack{1 \leq p < n \\ p \text{ even}}} p \rho^p \right), \\ c_n &= \frac{1}{(1-\rho)^2} \sum_{\substack{3 \leq t \leq n \\ t \text{ odd}}} \frac{t-1}{2} (\rho^t - 2\rho^{n+1} + \rho^{2n+2-t}). \end{aligned} \quad (\text{D.2})$$

In particular $d_n \geq 2\rho^3 > 0$.

D.1. An exchange bound from the forced means. On the mixed coordinates introduce the positive matrix

$$(W_\rho)_{pq,rs} = \frac{1}{2} \rho^{(p+q+r+s)/2} \cos \frac{\pi(p+q-r-s)}{2}.$$

It is half the sum of the two rank-one matrices with vectors $\rho^{(p+q)/2} \cos(\pi(p+q)/2)$ and $\rho^{(p+q)/2} \sin(\pi(p+q)/2)$. The product-to-sum identity gives

$$(W_\rho)_{pr,qs} - (W_\rho)_{ps,qr} = -z_{pq} z_{rs}, \quad \langle \mathcal{R}(\Xi), W_\rho \rangle = -2z^T \Xi z.$$

The low-total parts of both vectors lie in the span of the forced means. Thus, writing $W_{\rho,\text{hi}}$ for the compression to totals $> n$,

$$\langle \mathcal{R}(\Xi), W_\rho \rangle = \langle \mathcal{R}(\Xi), W_{\rho,\text{hi}} \rangle.$$

Since $M_n^0, W_{\rho,\text{hi}} \succeq 0$, this implies $z^T \Xi z \geq -\frac{1}{2} \langle M, W_{\rho,\text{hi}} \rangle$. The mixed diagonal is fixed and satisfies $0 \leq M_{pq,pq} \leq 2pq$. Positive semidefiniteness and the bound on each entry of W_ρ therefore give

$$\langle M, W_{\rho,\text{hi}} \rangle \leq S^2/2, \quad S = \sum_{p+q>n} \sqrt{M_{pq,pq}} \rho^{(p+q)/2} \leq \frac{\sqrt{2} n^2 \rho^{(n+1)/2}}{1 - \sqrt{\rho}}.$$

There are at most n mixed coordinates on each total. Testing the minus pure sector in (D.1) now yields

$$\gamma_n \geq \frac{c_n - n^4 \rho^{n+1} / (1-\rho)^2}{d_n}. \quad (\text{D.3})$$

Indeed $z^T \Xi z \geq -S^2/4$ and, since $1 - \rho \leq 2(1 - \sqrt{\rho})$, $S^2/8 \leq n^4 \rho^{n+1} / (1-\rho)^2$. The estimate is uniform in every feasible Ξ, E, τ , so it also holds at the minimum budget.

D.2. Summing the scalar kernel. For an odd gap h , put $l = n - h$. Its block of J_n is $\min(p, r, l - p, l - r)$, $1 \leq p, r < l$, with factorization

$$\min(p, r, l - p, l - r) = \sum_{j=1}^{\lfloor l/2 \rfloor} \mathbf{1}_{j \leq p \leq l-j} \mathbf{1}_{j \leq r \leq l-j}.$$

Consequently

$$c_n = \sum_{h \text{ odd}} \rho^h \sum_{j=1}^{\lfloor (n-h)/2 \rfloor} \left(\sum_{p=j}^{n-h-j} \rho^p \right)^2.$$

Putting $t = h + 2j$ gives (D.2). The corresponding infinite sums are

$$d_\infty = \frac{2\rho^3(1+\rho^2)}{(1-\rho^2)^4}, \quad c_\infty = \frac{\rho^3(1+\rho)^2}{(1-\rho^2)^4}, \quad \frac{c_\infty}{d_\infty} = 1 - \frac{(1-\rho)^2}{2(1+\rho^2)}. \quad (\text{D.4})$$

They follow from the odd and even weighted geometric series.

For retained triples (h, p, r) , the lost coefficient is

$$\min(p, r) - \min(p, r, n - h - p, n - h - r) = (p + r + h - n)_+.$$

An omitted triple also has exponent $p + r + h > n$. There are at most t^2 triples of exponent t , each with lost coefficient at most t . For $s = 1 - \rho$ and $ns \geq 8$, it follows that

$$\begin{aligned} 0 \leq c_\infty - c_n &\leq \sum_{t>n} t^3 \rho^t \\ &\leq \rho^n \left(\frac{n^3}{s} + \frac{3n^2}{s^2} + \frac{6n}{s^3} + \frac{6}{s^4} \right) \leq \frac{2n^3 \rho^n}{s}. \end{aligned} \quad (\text{D.5})$$

Here the four geometric moment bounds follow from $\sum_{j \geq 1} j^a \rho^j \leq a! / s^{a+1}$ for $a = 0, 1, 2, 3$.

For $n \geq 128$, let $\ell_{\log} = \lceil \log_2 n \rceil$, $s = 8\ell_{\log}/n$, and $\rho = 1 - s$. Then $0 < s \leq 1/2$ and $\rho^n \leq e^{-8\ell_{\log}} \leq 2^{-8\ell_{\log}} \leq n^{-8}$. Using $c_n \geq 0$, $d_n \leq d_\infty$, and $d_n, d_\infty \geq 2\rho^3$, equations (D.3)–(D.5) give

$$\begin{aligned} \gamma_n &\geq 1 - \frac{s^2}{2} - \frac{8n^3 \rho^n}{s} - \frac{2n^4 \rho^n}{s^2} \\ &\geq 1 - \frac{32\ell_{\log}^2}{n^2} - \frac{1}{\ell_{\log} n^4} - \frac{1}{32\ell_{\log}^2 n^2} \geq 1 - \frac{33\ell_{\log}^2}{n^2}. \end{aligned}$$

For the last inequality, $\ell_{\log} \geq 7$ and $n \geq 128$ suffice. This proves the quantitative part of Theorem 9.2.

D.3. The uniform bounds. The mixed coordinate z_{11}^{mix} has diagonal zero, so positivity forces its row to vanish. In particular $\Xi_{12,12} = M_{11,22} = 0$. Since a pure four-form has zero diagonal and $(J_n)_{12,12} = 1$, the minus pure diagonal is $4\tau - 2$. Thus every feasible budget satisfies $\tau \geq 1/2$.

For the upper bound, $J_n \succeq 0$ by its interval factorization. In each gap block,

$$0 \leq \frac{(J_n)_{p,p+h;r,r+h}}{\sqrt{p(p+h)r(r+h)}} \leq \frac{1}{\max(p, r)}.$$

The weighted Schur test with weight $p^{-1/2}$ gives

$$\frac{1}{p} \sum_{r \leq p} r^{-1/2} + \sum_{r > p} r^{-3/2} \leq 4p^{-1/2},$$

by comparison with the integrals over $(0, p)$ and (p, ∞) . Taking absolute values of a vector before applying this entrywise bound proves $0 \preceq J_n \preceq 2D_n$. Therefore $\Xi = E = 0$ and $\tau = 4$ make $4D_n \pm 2J_n$ positive, and the mixed matrix is $M_n^0 \succeq 0$. This proves $\gamma_n \leq 4$ and completes the proof.

APPENDIX E. FORMAL AND COMPUTATIONAL VERIFICATION

E.1. Formal results. The negative conclusion of Theorem 1.1 is formally proved in Lean 4 with Mathlib, unconditionally. The formal project defines the literal quartic as `ToeplitzSOS.toeplitzBW` and uses the predicate `ToeplitzSOS.IsSumSqHomQuad` for finite sums of real homogeneous quadratic squares. The theorem `ToeplitzSOS.Negative.sharpNegativeResolution` proves `SharpNegativeResolution`, namely non-SOS for every $N \geq 2^{9961475}$. The declarations `ToeplitzSOS.Negative.sharp_negative_explicit` and `ToeplitzSOS.Negative.not_isSumSqHomQuad_of_large_order` give the expanded quadratic quantifiers and the direct predicate form, respectively. The declaration `ToeplitzSOS.Negative.not_MI15` refutes the registered catalogue statement. The classical nonnegativity assertion is proved in Section 1.3. The formal positive range is $2 \leq N \leq 20$.

The proof composes `ToeplitzSOS.Negative.cornerGram_of_sos`, which extracts the complete corner data from an arbitrary SOS of the literal polynomial, with `ToeplitzSOS.Negative.no_cornerGram`, which excludes those data at the selected depth. The analytic inputs are supplied by the proved `ToeplitzSOS.Negative.Analytic.analyticInputs`. They include `ToeplitzSOS.Negative.Analytic.baselineTail`, the mixed correction bound, and the two-rectangle comparison; their coefficient, complete-mean, and majorant hypotheses are proved for the corner data extracted from an arbitrary sum of squares. The integer witness is checked by reduction in the Lean kernel in `ToeplitzSOS.Negative.WitnessCertificate`, and `ToeplitzSOS.Negative.Witness.tangent_pairing_lt` proves its negative kernel pairing.

The formal final budget uses the exact witness mass bound $\|c\|_1^2 \leq 3 \cdot 2^{58}$. At the selected scale the realigned entries are at most 2^{-18} , so their contribution is at most $3 \cdot 2^{58} \cdot 2^{-18} = 3 \cdot 2^{40} < 2^{42}$; the finite-to-tangent contribution is at most $3/4$. Thus the total error is less than 2^{42} without the convexity step used for the coarser mass bound in Corollary 1.2.

The formal proofs use only the audited standard axioms `propext`, `Classical.choice`, and `Quot.sound`.

E.2. The result map. The table lists the numbered results; declaration names are relative to the namespace `ToeplitzSOS`. The last column names the exact computation behind a result, identified in the accompanying verification map; “Written proof” means that the argument in this paper uses no computation. The Lean declarations give formal proofs within the scopes stated in the table. Some intermediate results are organized differently in the formal proof: it uses polarization in place of the full relation-space classification and closes the finite contradiction at the selected scale with the budget just described. The exact positive chain proves orders through 50, and its Lean scope is as shown; rational Gram certificates suffice for the real problem (Section 8).

Result	Lean declaration	Exact check
Theorem 1.1 Negative answer	Negative.sharpNegativeResolution Negative.sharp_negative_explicit Unconditional non-SOS assertion; nonnegativity is classical.	negative.py witness.py
Corollary 1.2 Explicit threshold	Negative.cornerGram_impossible Negative.not_isSumSqHomQuad_of_large_order The selected corner and every larger ambient order.	negative.py witness.py
Lemma 3.1 The complete quadratic frame	Negative.exists_alternating_sos Negative.alternating_eq_wedge_sum SOS-to-wedge reduction; the full relation-space classification is replaced by polarization.	Written proof
Lemma 3.2 Stabilization and the complete normal form	Negative.outerCorner_eq_baselinePolynomial Negative.cornerGram_of_sos Complete literal corner extraction from arbitrary SOS data.	Written proof
Lemma 4.1 Forced complete means	Negative.corner_sos_low_means Negative.baselineMixedEntry_low_column Complete low means; baseline positivity is given by the written Laplacian proof.	Written proof
Lemma 4.2 Global bound and the repeated-node identity	Negative.CornerGram.repeated_diagonal Negative.referenceQ_conjugate Negative.CornerGram.plusVector_global Kernel identities and a sufficient global majorant; the formal radial estimate uses a looser constant.	Written proof
Lemma 4.3 Uniform finite tails	Negative.Analytic.baselineTail Negative.CornerGram.repeated_tail_bound Both baseline truncations and the combined repeated-node error; the corner extraction supplies the coefficient and mean hypotheses.	Written proof
Lemma 5.1 Uniform improvement of the pure kernel	Negative.Analytic.twoRectangles Negative.CornerGram.realigned_bound General comparison; Gram majorants and the realigned witness entries are discharged at the selected scale.	Written proof
Lemma 6.1 Exact negative pairing	Negative.Witness.tangent_pairing_lt Negative.Witness.fullUpper_lt Exact integer certificate checked by Lean kernel reduction.	negative.py witness.py
Theorem 7.1 Uniform finite-scale obstruction	Negative.finite_separation Negative.no_feasible_evaluation Formal separation at the selected scale; the uniform budget is the written version.	negative.py witness.py
Proposition 8.1 Exact positive certificates	toeplitzBW_isSumSq_of_le_20 Orders 2 through 20 only.	positive.py
Theorem 9.1 Exact exchange criterion	Not formalized	Written proof
Theorem 9.2 Criticality of the pure budget	Not formalized	Written proof
Corollary 10.1 Failure of the all-order increment condition	Not formalized	Written proof

Result	Lean declaration	Exact check
Remark 11.1 An expectation at moderate orders	Not formalized	None (unproved remark)

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